National Park Demand Signals and a Cloud Lakehouse

Course project proposal · October 2026
Team: Chujie Wang, Fan Du, You Fu, Tianyang Zhang

Problem and users

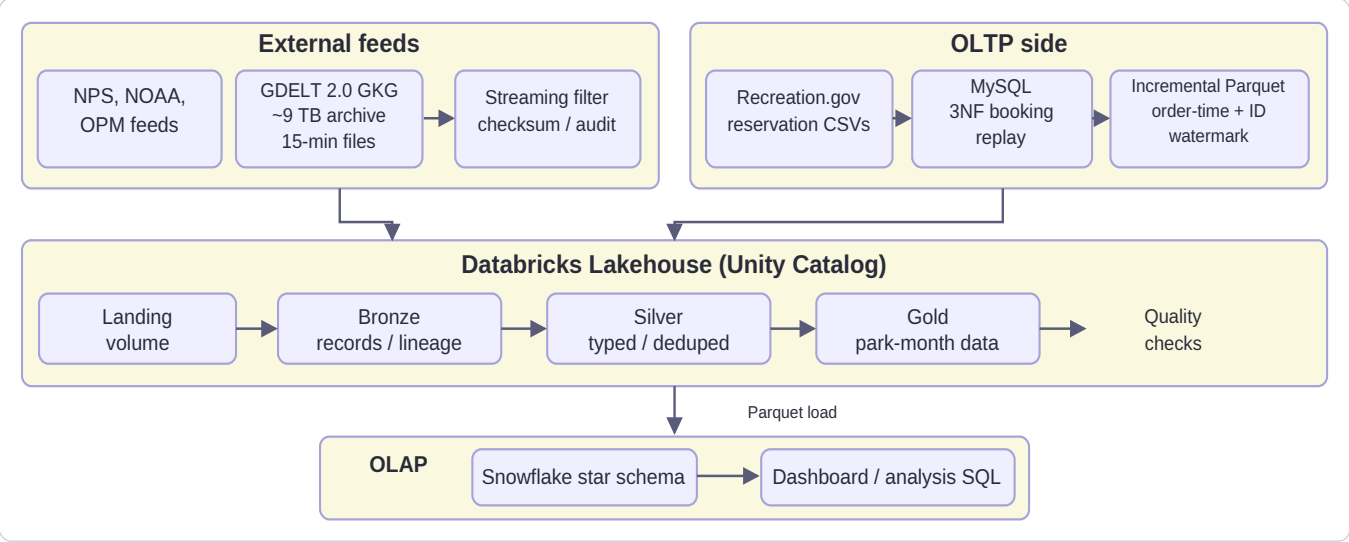
We will ask whether reservation activity and news coverage help explain or anticipate monthly visits to Yellowstone, Yosemite, Grand Canyon, Zion, and Great Smoky Mountains. Park managers need staffing and shuttle plans; nearby businesses need capacity plans. We compare each signal with a seasonality-only baseline on National Park Service (NPS) recreation visits for 2015–2025: 5 parks × 132 months = 660 park-months.

Datasets, rights, and the 5 TB data rule

Dataset	Location and role	Rights
GDELT 2.0 Global Knowledge Graph (GKG)	data.gdeltproject.org/gdeltv2; news coverage, **scale driver**	Free for any use with citation (terms)
Recreation.gov reservations	ridb.recreation.gov/download; booking events for MySQL	RIDB API Access Agreement; we credit RIDB and publish aggregates only
NPS visits, NOAA weather, OPM holidays	NPS, NOAA, OPM; outcome and context	Public domain; already landed

How the project meets the data rule. The GKG is **about 9 TB uncompressed** (2.88 TB zipped): about 740 million article records in 402,046 files, and **a new file arrives every 15 minutes** since February 2015. That is more than a laptop can store or hold in memory, so we never keep the raw archive. A resumable worker downloads each file, verifies its MD5 checksum, keeps the records naming one of the five parks, counts all records per day as a denominator, and discards the file. Only matched records and daily counts enter the lakehouse. The reservations add the relational workload: FY2024 alone has 9.78 million bookings, 1.29 million in our parks. None of the data relates to Sakila or ClassicModels.

Target architecture



The loader replays reservations into MySQL as idempotent transactions; an extract job exports rows past an order-time and ID watermark as Parquet. Databricks builds Bronze, Silver, and Gold tables with quality checks, and Gold loads into a Snowflake star schema. The small-feed ingestion and Bronze/Silver code already exist.

Two riskiest assumptions and Week 8 tests

1. We can scan the full GDELT archive on schedule and match parks precisely. Sampled files download at about 10 MB/s, so one connection needs about 3.3 days just to download the 2.88 TB, and the file list is missing 7,286 fifteen-minute slots, including a 17-day outage in June–July 2025. Matching is noisy: in 24 sample files the bare word "Zion" appeared in 25 records, and only 5 referred to the park. **Week 8 test (Fan, Chujie):** process one week (672 files) end to end and hand-check 100 matched records. Pass if every checksum verifies, a rerun reproduces the counts, at least 90% of checked matches are about the park, and the projected full scan takes 7 days or less. If not, the four members split the years across their own machines.

2. Reservations are usable leading signals. The reservation files change layout across years (56 columns in FY2015–FY2017, 35 from FY2021, and the park-name field moves in FY2019), and booked slots can reflect permit supply rather than demand. **Week 8 test (You Fu, Tianyang):** You Fu loads FY2017 and FY2024 into MySQL, maps facilities to the five parks, and checks keys and row counts against the source files. Tianyang computes the reservations on the books at the start of each month from the FY2015–FY2025 files and compares a seasonal baseline with and without them on held-out months from January 2024 to September 2025, training without the 2020 closures. Pass if row counts match and reservations lower held-out error in at least three parks; otherwise the result is descriptive, not a forecast. News coverage joins the comparison after the full scan.

Team roles

Member	Primary responsibility
Chujie Wang	Integration lead; NPS/NOAA/OPM feeds, Databricks Silver/Gold, quality checks
Fan Du	GDELT streaming filter, park matching, throughput benchmark, ingestion audit
You Fu	Recreation.gov mapping, MySQL OLTP model, replay loader, incremental extract
Tianyang Zhang	Snowflake star schema, dashboard, held-out evaluation